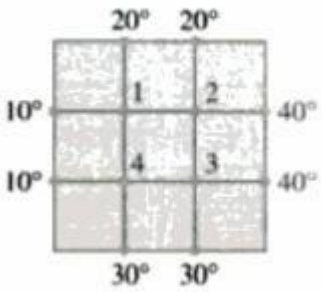
UE25MA242A: Mathematical Foundation for AI and Data Science

Question Bank: Unit 1

1.	Explain the row approach to solve the system $2x - 3y = 3, x + 3y = 6$ with a neat diagram. What happens to the solution when the second equation isreplacedby $x + 3y = -6$? **Answer**: Solution is $x = 3, y = 1$; New solution is $x = -1, y = -5/3$.
2.	Explain the column approach to solve the system $x - 2y = 3, 2x + y = 1$ with a neat diagram. What happens to the solution if the second equation is replaced by $2x - 4y = 5$? **Answer:** $x = 1, y = -1$. If the second equation is replaced by $2x - 4y = 5$ then the system is singular.
3	Solve the following systems of equations using Gaussian elimination: (i) $2x + y + 3z = 1, 2x + 6y + 8z = 3, 6x + 8y + 18z = 5$ **Answer**: $x = 3/10, y = 2/5, z = 0$ (ii) $x + 2y - z = 6,\ 2x + y + z = 3,\ x - y + z = -2$ **Answer**: $x = 1, y = 2, z = -1$ (iii) $3x + y - 6z = -10, 2x + y - 5z = -8, 6x - 3y + 3z = 0$ **Answer**: $x = k - 2, y = 3k - 4, z = k$ where k is a scalar (iv) $x + z = 1, x + y + z = 2, x - y + z = 1$. What if the right hand side is $(1, 2, 0)$? **Answer** : Inconsistent system ; $x = 1 - k, y = 1, z = k, k \in R$
4	Investigate the values of λ and μ such that $$x + 3y + 5z = 9$$ $$x - y + 2z = 1$$ $$2x + 2y + \lambda z = \mu$$ has (i) unique solution (ii) infinitely many solution (iii) no solution **Answer:** (i) unique solution when $\lambda \neq 7$ (ii) $\lambda = 7,\ \mu = 10$ (iii) λ should be equal to 7 $(r(A) = 2)$ and $\mu - 10 \neq 0$ i.e., $\mu \neq 10$ $(r(A:b) = 3)$.
5.	Determine the values of a and b for which the system of equations $x + y + az = 2b$, $x + 3y + (2 + 2a)z = 7b, 3x + y + (3 + 3a)z = 11b$willhave (i) unique nontrivialsolution (ii)trivialsolution (iii)nosolution (iv) infinity ofsolutions. **Answer** : (i) $a \neq -5$ and any b (ii) $a \neq -5$ and $b = 0$ (iii) $a = -5$ and $b \neq 0$ (iv) $a = -5$and $b = 0$.
6.	Let $A = \begin{bmatrix} 3 & -6 & 2 & -1 \\ -2 & 4 & 1 & 3 \\ 0 & 0 & 1 & 1 \\ 1 & -2 & 1 & 0 \end{bmatrix}$ and $b = (b_1, b_2, b_3, b_4)$. Use the method of Gaussian Elimination to find a condition on thecomponentsof b so that thesystem$Ax = b$ is consistent. When $b = (2, 1, 1, 1)$, if $(x, 0, 0, 1)$ is a solution of the system $Ax = b$ findx. **Answer**:$7b_4 - b_2 - 3b_1 = 0$ and $7b_3 - 3b_2 - 2b_1 = 0$. When $b = (2, 1, 1, 1)$then $3x - 6y + 2z - t = 2, z + t = 1$, Solving we get $x = 1$.
7.	Determine the equation of the polynomial $y = f(x)$ of degree 2 whose graph passes through thepoints $(1, 6)$, $(2, 3)$ and $(3, 2)$. **Answer** :$y = 11 - 6x + x^2$.
8.	Which three matrices E_{21}, E_{31}, E_{32} put A into a triangular form $A = \begin{bmatrix} 1 & 1 & 0 \\ 4 & 6 & 1 \\ -2 & 2 & 0 \end{bmatrix}$. Multiply those E's to get one matrix M thatdoeselimination $MA = U$.

	Answer : $M = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 10 & -2 & 1 \end{bmatrix}$
9.	Write down the elementary matrices E , F , G associated with the system of equations $2u + v + 3w = -1$, $4u + v + 7w = 5$, $-6u - 2v - 12w = -2$. Also find the LU decomposition of A. **Answer :** $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -3 & -1 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 3 \\ 0 & -1 & 1 \\ 0 & 0 & -2 \end{bmatrix}$
10.	Find L and U for the matrix $A = \begin{bmatrix} 1 & -1 & 0 & 3 \\ 2 & -2 & 0 & 8 \\ -1 & -1 & 4 & -2 \\ -2 & -2 & 6 & -3 \end{bmatrix}$. Write down the permutation matrices, if any, used in the process of elimination. **Answer**: $L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ -2 & 2 & 1 & 0 \\ 2 & 0 & 0 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & -1 & 0 & 3 \\ 0 & -2 & 4 & 1 \\ 0 & 0 & -2 & 1 \\ 0 & 0 & 0 & 2 \end{bmatrix}$ The permutation matrices used are P_{23} and P_{34}
11.	Find LU and LDU factorization for $A = \begin{bmatrix} 3 & 1 & 2 \\ 2 & -3 & -1 \\ 1 & 2 & 1 \end{bmatrix}$ **Answer:** $A = LDU = \begin{bmatrix} 1 & 0 & 0 \\ \frac{2}{3} & 1 & 0 \\ \frac{1}{3} & \frac{-5}{11} & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 & 0 \\ 0 & -\frac{11}{3} & 0 \\ 0 & 0 & -\frac{24}{33} \end{bmatrix} \begin{bmatrix} 1 & \frac{1}{3} & \frac{2}{3} \\ 0 & 1 & \frac{7}{11} \\ 0 & 0 & 1 \end{bmatrix}$
12.	Find $PA = LDU$ factorization for $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 2 & 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 1 \end{bmatrix}$ **Answer**: $PA = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ $PB = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$
13.	Find the symmetric factorization of A in the form $L\,D\,L^T$ and find conditions on a, b, c , d to get $A = LU$ with four pivots.$A = \begin{bmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ a & b & c & d \end{bmatrix}$
	Suppose A is a 4 x 4 identity matrix except for a vector v in column 2: Factor A into LU assuming

14.	$v_2 \neq 0. A = \begin{bmatrix} 1 & v_1 & 0 & 0 \\ 0 & v_2 & 0 & 0 \\ 0 & v_3 & 1 & 0 \\ 0 & v_4 & 0 & 1 \end{bmatrix}$
15.	Use the Gauss – Jordan method to invert the followingmatrices $.i. A = \begin{bmatrix} 1 & 2 & -1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{bmatrix}$ $ii\ A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$ $iii\ A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$ **Answer**: i. $A^{-1} = \frac{1}{14}\begin{bmatrix} 3 & -1 & 5 \\ 5 & 3 & -1 \\ -1 & 5 & 3 \end{bmatrix}$ ii.$A^{-1} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$ iii.$A = \begin{bmatrix} -2/8 & 4/8 & 2/8 \\ 3/8 & -2/8 & 1/8 \\ 7/8 & -2/8 & -3/8 \end{bmatrix}$
16.	Producing x trucks and y planes requires $x + 50\ y$ tons of steel, $40\ x + 1000\ y$ pounds of rubber and $2x + 50\ y$months of labor. If the unit costs u, v, w are \$ 700 per ton, \$ 3 per pound and \$ 3000 per month, what are the values of one truck and one plane? **Answer**: 6820 , 188000
17.	Assume that the plate shown in the figure represents a cross section of a metal beam with negligible heat flow in the direction perpendicular to the plate. Let T_1, T_2, T_3 and T_4 denote the temperatures at the four interior nodes of the mesh. The temperature at a node is approximately equal to the average of the four nearest nodes – to the right, left, above and below. For instance $T_1 = (10 + 20 + T_2 + T_4) / 4$ or $4\ T_1 - T_2 - T_4 = 30$. Write a system of 4 equations whose solution gives estimates for the temperaturesT_1, T_2 , $T_3$and T_4.Hence find its solution. 
18.	Propane is a common gas used for cooking and home heating. Each molecule of propane is comprised of 3 atoms of carbon, and 8 atoms of hydrogen written as C_3H_8. When propane burns, it combines with oxygen gas O_2 to form carbon dioxide CO_2 and water H_2O. Balance the chemical equation $C_3H_8 + O_2 \rightarrow CO_2 + H_2O$ that describes this process. **Answer**: $2\ C_3H_8 + 10\ O_2 \rightarrow 6\ CO_2 + 8\ H_2O$